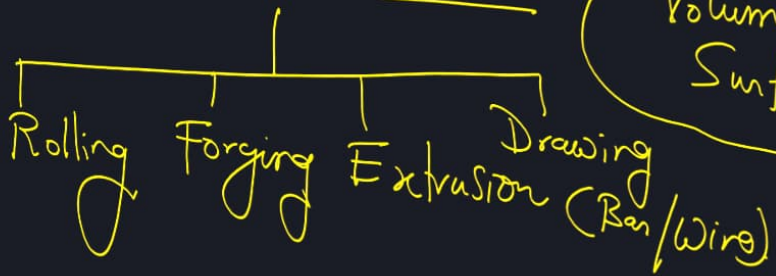


Date: - 07/01/2026

Manufacturing Process.

Metal Forming.

Bulk Deformation



Bulk Deformation Process

Volume is more than Surface Area.

Metal Forming

Sheet Metal Working

Surface Area is more than the Volume.

Hot Working

① Defⁿ:

➤ Recrystallization Temperature.

$$T_{\text{Recryst}} \approx 0.5 T_m \sim 0.75 T_m$$

T_m = melting point of the metal
in Absolute Scale.

Cold Working

① Much below Recrystallization Temperature.

Hot Working

② Surface Finish.

Poor

③ Dimensional Accuracy.

Poor

④ Strain Hardening
Minimum.

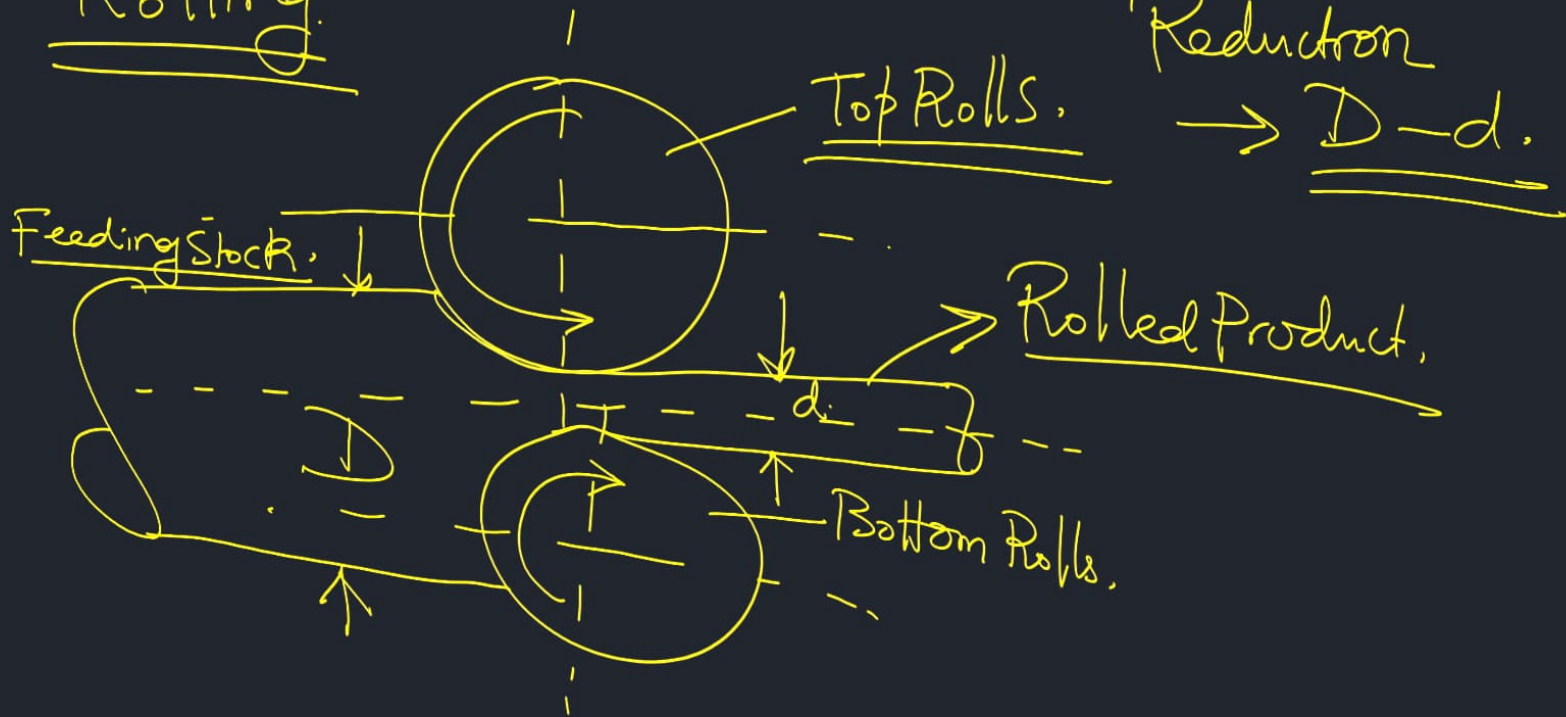
Cold Working

② Better

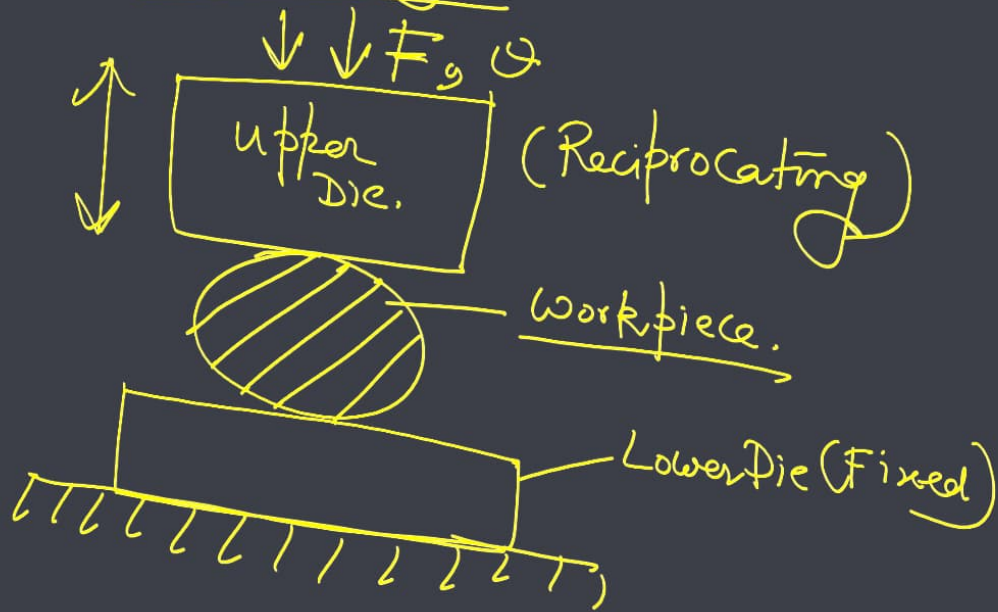
③ Better.

④ Substantial.

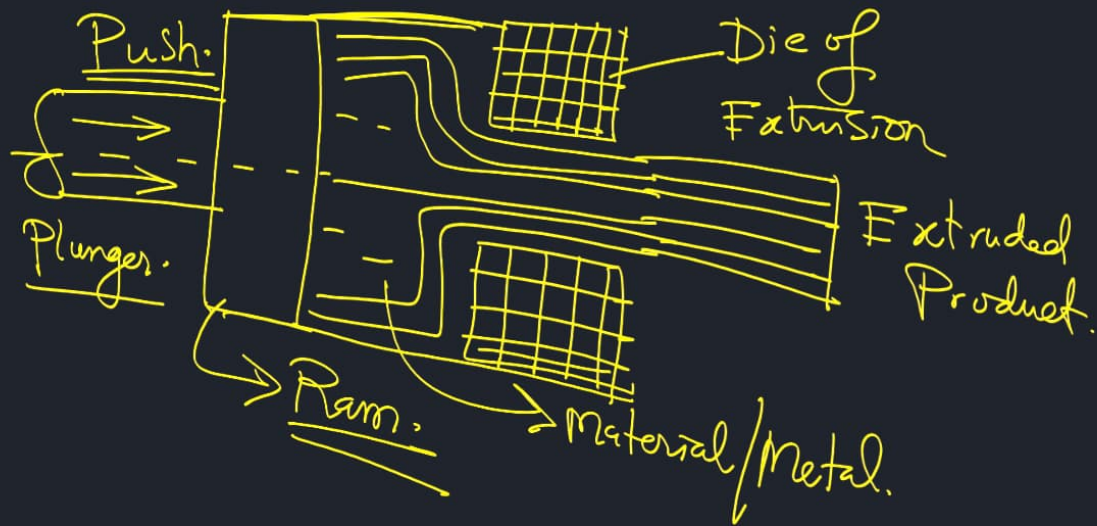
Rolling



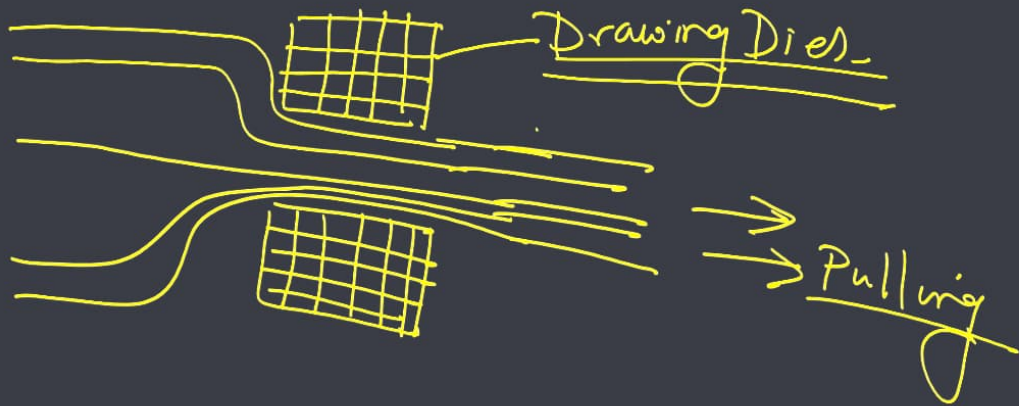
Forging



Extrusion.



Drawing (Wire/Bar)



Date: - 09/01/2026

Metal Forming

$$\text{True Stress} = \frac{\text{Force}}{\text{Instantaneous Cross Section}}$$

$$\text{True Strain } \epsilon = \ln \frac{L}{L_0} \quad \text{Cross Section}$$

$L_0 = \text{original.}$

Material Behaviour in Metal Forming

Flow Stress $\sigma_f = K \epsilon^n$

$\sigma_f =$ in MPa, $K =$ Strength Coefficient in MPa

$\epsilon =$ True Strain (Dimensionless)

$n =$ Strain Hardening Exponent

$$\sigma_f = \frac{K \epsilon^n}{1+n}$$

Problem.

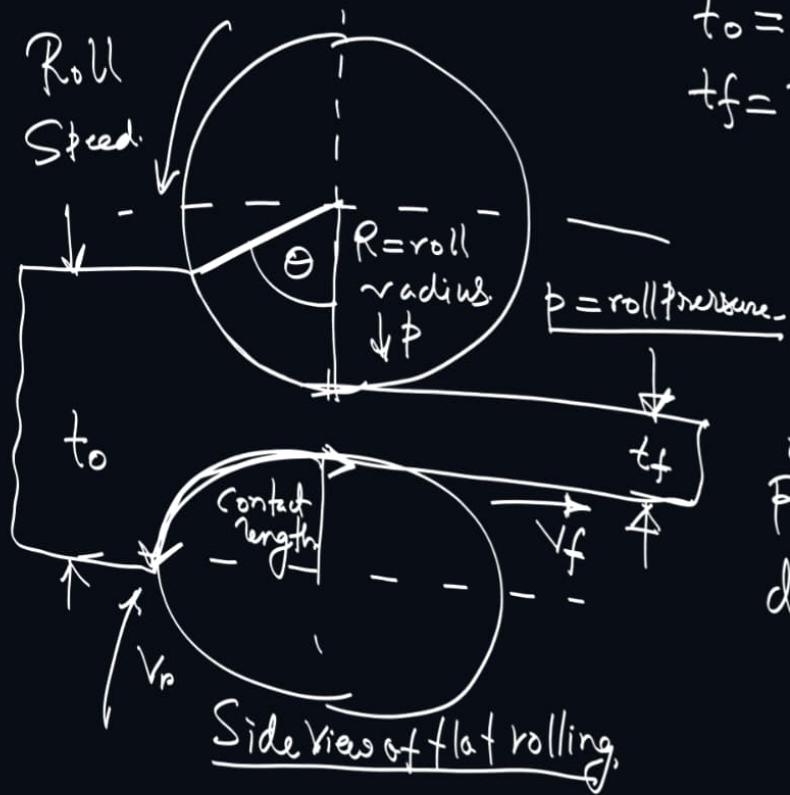
$$\underline{\underline{K = ???, n = ???}}$$

In a tensile test two pairs of values of Stress Strain are measured for a specific metal specimen.

i) True Stress (Flow Stress) = 217 MPa
True Strain = 0.35

ii) Flow Stress = 259 MPa
True Strain = 0.68

A 300 mm wide Strip 25 mm thick is fed thro a rolling mill with two powered rolls each of radius = 250 mm. The work thickness is reduced to 22 mm in one pass and the roll speed is 50 rev/minute. The work material has a flow curve defined by $K = 275 \text{ MPa}$, $n = 0.15$. $\mu = 0.12$. Determine if the friction is sufficient to permit the rolling operation to be accomplished. If so Calculate the roll force, torque & horse power.



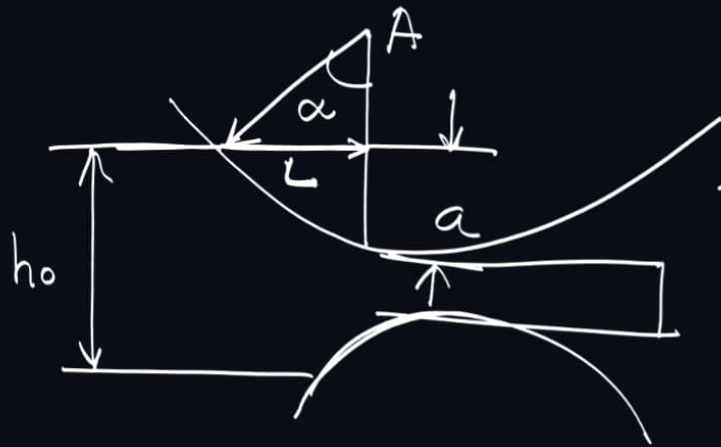
$t_o = \text{original thickness}$
 $t_f = \text{Final thickness.}$

$$d_{\max} = \mu^2 R$$

$d_{\max} = \text{Maximum draft}$
 $\mu = \text{Coefficient of friction between the roll and workpiece material.}$
 $R = \text{Roll Radius.}$

$$d_{\max} = 25 \text{ mm} - 22 \text{ mm} = 3 \text{ mm.}$$

Contact Length.



So Contact Length $(L) = \sqrt{R \times \Delta h}$

$$\begin{aligned} R^2 &= L^2 + (R-a)^2 \\ \textcircled{L^2} &= R^2 - (R-a)^2 \\ &= R^2 - (R^2 + a^2 - 2Ra) \\ &= 2Ra - \textcircled{a^2} \text{ negligible.} \\ &\approx \textcircled{2Ra} \\ &= \textcircled{R \times \Delta h.} \end{aligned}$$

$$\underline{\underline{L}} = \sqrt{R \times (t_o - t_f)}$$

L = Contact Length

t_o = original thickness

t_f = Final thickness.

$$\underline{\underline{L}} = \sqrt{250 \times (25 - 22)} \\ = 27.4 \text{ mm.}$$

For Rolling we have to
Consider average Flow Stress

$$\overline{y_f} = \frac{K \epsilon^n}{1+n}$$

$$\epsilon = \ln \frac{25}{22} = 0.128$$

$$\overline{y_f} = \frac{250 \times (0.128)^{0.15}}{1 + 0.15} \\ \text{(Rolling)} \approx 175.7 \text{ MPa}$$